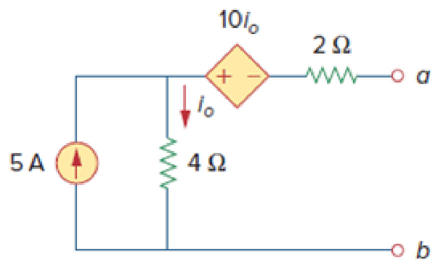
Chapter 4, Problem 48P

Problem

Determine the Norton equivalent at terminals *a-b* for the circuit in Fig. 4.115.

Figure. 4.115:



Step-by-step solution

Step 1 of 6

Refer to the circuit diagram in Figure 4.115 in the text book.

Consider the circuit diagram shown in Figure 1 to calculate the Thevenin's voltage.

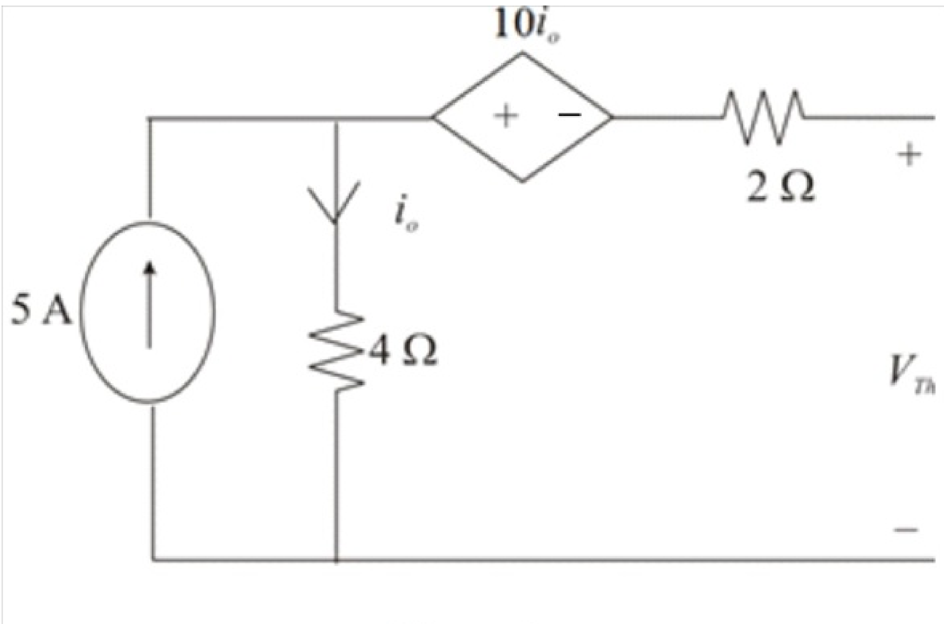


Figure 1

Step 2 of 6

The value of current i_o is 5 A.

Calculate the value of the Thevenin's voltage V_{Th}.

$$\begin{aligned} V_{Th} &= i_o(4) - 10i_o \\ &= -6i_o \end{aligned}$$

Substitute 5 A for i_o.

$$\begin{aligned} V_{Th} &= -6(5) \\ &= -30\text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is $\boxed{-30\text{ V}}$.

Step 3 of 6

Consider the circuit diagram shown in Figure 2 to calculate the short circuit current.

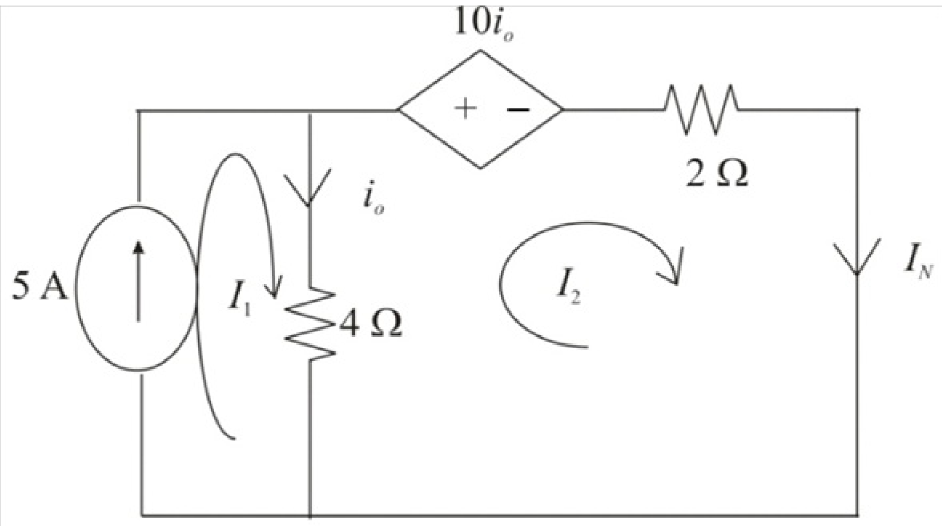


Figure 2

Step 4 of 6

The value of the current I_1 is 5 A.

The value of the current i_o is,

$$i_o = I_1 - I_2$$

Apply Kirchhoff's Voltage Law in loop 2.

$$10i_o + (I_2 - I_1)(4) + I_2(2) = 0$$

Substitute $I_1 - I_2$ for i_o.

$$\begin{aligned}10(I_1 - I_2) + (I_2 - I_1)(4) + I_2(2) &= 0\\ I_1(10-4) - I_2(10-4-2) &= 0\\ I_1(6) - I_2(4) &= 0\end{aligned}$$

Substitute 5 A for I_1.

$$\begin{aligned}(5)(6) - I_2(4) &= 0\\ I_2(4) &= 30\\ I_2 &= \frac{30}{4}\\ I_2 &= 7.5\text{ A}\end{aligned}$$

Here, I_2 represents the Norton's current I_N.

$$I_N = 7.5\text{ A}$$

Therefore, the value of the Norton's current I_N is, $\boxed{7.5\text{ A}}$.

Step 5 of 6

Calculate the value of the value of the Norton's resistance.

$$R_N = \left|\frac{V_{Th}}{I_N}\right|$$

Substitute 7.5 A for I_N and -30 V for V_{Th}.

$$R_N = \left|\frac{-30\text{ V}}{7.5\text{ A}}\right|$$
$$= 4\ \Omega$$

Therefore, the value of the Norton's resistance is $\boxed{4\ \Omega}$.

Step 6 of 6

The Norton's equivalent circuit diagram is shown in Figure 3.

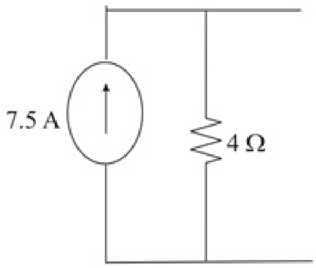


Figure 3

Therefore, the Norton's equivalent circuit is shown in Figure 3.